

Ryan Peach

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Objective: Senior infrastructure engineer with a decade of experience running large-scale distributed systems on AWS and Kubernetes.

EDUCATION

Georgia Tech	Atlanta, GA
<i>MS in Computer Science, Machine Learning Specialty. Computer Vision & Robotics Equiv. Coursework</i>	<i>2015 – 2018</i>
Western Kentucky University	Bowling Green, KY
<i>BS in Electrical Engineering, Minor in Mathematics</i>	<i>2010 – 2014</i>

MAJOR SKILLS

<i>Docker, Kubernetes, AWS</i>	<i>Expert, 2017-Present</i>
<i>CI/CD & DevOps: GitHub Actions, Jenkins, GitOps, Helm</i>	<i>Expert, 2017-Present</i>
<i>Infrastructure as Code: Terraform, CloudFormation</i>	<i>Expert, 2017-Present</i>
<i>Python Scientific Stack, Data Science</i>	<i>Expert, 2013-Present</i>
<i>Machine Learning: Deep Learning, Reinforcement Learning</i>	<i>MS, 2015-Present</i>
<i>Rust Language</i>	<i>Advanced, 2019-Present</i>

EXPERIENCE

Senior Principal Engineer	2023-2026
<i>Ansys / Synopsys</i>	<i>Atlanta, GA</i>

Architecture

- Architected globally available infrastructure spanning multiple regions: global databases, failover clusters, edge services, and read replicas to ensure low-latency access and high availability worldwide.
- Extended Kubernetes platform from cloud-only to hybrid on-prem and cloud environments, enabling deployment flexibility across enterprise customer sites.
- Established enterprise Terraform practices: standardized module structure, remote state management, and onboarding patterns for teams new to IaC.
- Implemented drift detection, policy standards, and Kyverno admission controls across Kubernetes clusters to enforce compliance and security posture at scale.
- Designed and maintained authoritative infrastructure diagrams covering dozens of interoperating services; trained management on enterprise infrastructure model.
- Designed GraphQL, gRPC, and OpenAPI APIs for core backend services, enabling independent service deployments.
- Built a source-of-truth data pipeline extracting state from Kubernetes, landing raw data in S3 for analysis and simultaneously writing to DynamoDB for low-latency operational reads.

Large Language Models

- Participated in early open source efforts with LangChain, LlamaIndex, etc. Contributing code to repos.
- Advised on company LLM projects across multiple teams, attended conferences on LLM research

Director of Cloud DevOps	2021-2023
<i>OnScale</i>	<i>Atlanta, GA</i>

Engineering Leadership

- Organized teams around blast radius boundaries — each team owned a bounded set of services with defined failure domains and on-call scope.
- Led HPC, Infrastructure, and ML teams through the Ansys acquisition, transitioning all to new management upon integration before moving to technical track.

Architecture

- Executed a strangler fig migration from a monolithic API to a microservices architecture, enabling independent deployments and clear team ownership.

- Applied Domain-Driven Design to decompose the platform into bounded contexts; established explicit API contracts between services enforced via OpenAPI specs.
- Authored and maintained C4 architecture diagrams as the authoritative platform reference across the engineering org.
- Managed custom Kubernetes controller orchestrating simulation jobs on high-performance compute (HPC) infrastructure.
- Mandated GitOps workflows org-wide; overhauled CI/CD pipelines to enforce branch protection, automated testing, and deployment gates.
- Stood up full observability stack (metrics, distributed tracing, structured logging) and led FinOps practice — reducing cloud spend while improving incident response.
- Built Plotly dashboards surfacing key platform and business metrics, giving engineering and leadership real-time visibility into system health and usage.

ML Infrastructure

- Architected end-to-end ML pipeline on AWS: ETL ingestion, ECS-based training, and automated periodic retraining — enabling a customer-facing simulation cost estimator.
- Owned cloud resource allocation and cost governance for the ML platform, balancing training throughput against AWS spend.

Machine Learning Researcher - Lead Architect

2017 – 2021

Keysight Technologies

Atlanta, GA

Architecture

- Built CloudFormation stacks using ECS and Lambda that spun up thousands of Docker containers to train, tune, and select optimal SVG models. Weeks of compute completed in an hour regardless of data scale.
- Lead the team in code reviews, managed all the DevOps for the team.
- Designed PC cluster hardware; managed on-prem Linux, Spark, PostgreSQL, MongoDB, and Kubernetes.
- Built ETL pipelines on AWS ingesting data into Redshift via S3 and EMR/Spark; worked with Glue for catalog and job orchestration.

Machine Learning

- Independently investigated almost a dozen potential ML applications at the company, providing preliminary results and feasibility studies.
- Wrote approximately 7 internal publications on patentable inventions; gave multiple company-wide presentations on findings.
- Implemented large-scale hyperparameter tuning on AWS: some custom-built, others via SageMaker.
- Applied Reinforcement Learning to multiple novel problem domains using Monte Carlo Tree Search and Q Learning.
- Built data visualizations and interactive dashboards in Plotly, Dash, Tensorboard, Matplotlib, Altair, and Jupyter.

Leadership

- Promoted from Developer to Researcher within only 1 year due to outstanding performance. Ranked in the top 10% of performers every year at Keysight.
- Interviewed over a dozen candidates within 3 years.
- Onboarded 3 coworkers in Python best practices and professional software design principles.
- Maintained coding standards, code review, and planning of development milestones.

TECHNICAL SKILLS

Primary Languages: Python, Rust, Bash/Unix, PostgreSQL, Terraform, CloudFormation

Additional Known Languages: Golang, C/C++, C#, Java, Spark/Scala, Cypher

Technologies: Linux, Git, Docker, Kubernetes, ECS, Kyverno, OPA, Supabase, Redis, MongoDB, DynamoDB, Neo4j, GitHub Actions, AWS, Datadog, VIM, Regex